

Unit Outline

STAT2005 (V.1) Computer Simulation Semester 1, 2026

Unit study package number:	STAT2005
Mode of study:	Internal
Tuition pattern summary:	Note: For any specific variations to this tuition pattern and for precise information refer to the Learning Activities section. Lecture: 1 x 2 Hours Weekly Workshop: 1 x 2 Hours Weekly This unit does not have a fieldwork component.
Credit value:	25
Pre-requisite units:	(COMP1005 (v.0) Fundamentals of Programming or any previous version OR COMP1007 (v.0) Programming Design and Implementation or any previous version) AND (STAT1002 (v.0) Statistical Data Analysis or any previous version OR STAT1005 (v.0) Introduction to Probability and Data Analysis or any previous version)
Co-requisite units:	Nil
Anti-requisite units:	Nil
Result type:	Grade/Mark
Approved incidental fees:	Information about approved incidental fees can be obtained from our website. Visit https://www.curtin.edu.au/students/essentials/fees/understanding-your-fees for details.
Unit coordinator:	Name: Ms Sam Wiwatanapataphee Phone: 08 9266 Email: 275287a@curtin.edu.au Location Building: 314 - Room: 464 Consult: Wed - Fri, 2-5 pm
Teaching Staff:	Name: Dr Sam Wiwatanapataphee Phone: 08 9266 Email: 275287a@curtin.edu.au Location Building: 314 - Room: 464
Administrative contact:	Name: Curtin Connect Phone: 1300 222 888 Email or Website: https://students.connect.curtin.edu.au/app/ask Location Building: 120 - Room: N/A
Learning Management System:	Blackboard

Acknowledgement of Country

Curtin University acknowledges Aboriginal and Torres Strait Islander people, the First Peoples of this place we call Australia, and recognise them for caring for Country for more than 60,000 years. We are honoured and grateful for the privilege to maintain campuses operating in Boorloo (Perth) and Karlkurla (Kalgoorlie) in Australia. We pay our respects to Elders past and present as Custodians and Owners of these lands. We recognise their deep knowledge and their cultural, spiritual and educational practices, and aspire to learn and teach in partnership with them. Curtin also acknowledges First Nations peoples connected with our global campuses. We are committed to working in partnership with all Custodians and Owners to strengthen and embed First Nations' voices and perspectives in our decision-making, now and into the future.

Syllabus

This unit introduces the foundation of simulation techniques and their applications in a chosen area. The unit introduces the theoretical foundation of simulation with a focus on its scope and limitation. Several tools and techniques for designing and implementing simulation procedures on various computing platforms will also be introduced. Students will then have the opportunity to choose an area of focus where they can apply the principles of simulation on solving discipline specific problems.

Introduction

This course explores the pivotal role of modelling and simulation as essential tools for understanding, analysing, and solving non-standard real-world problems. Advances in modern computing have significantly expanded our ability to address increasingly complex problems through simulation, often outperforming traditional analytical methods in terms of efficiency and flexibility.

The unit provides a comprehensive introduction to the design and implementation of fundamental simulation studies, with applications drawn from a wide range of disciplines. It begins by examining the scope and limitations of computer simulations, followed by core topics including probability distributions, statistical inference, and simulation methodologies. Students are introduced to techniques for simulating discrete and continuous random variables, both univariate and multivariate, using classical and Bayesian approaches. More advanced simulation algorithms are then presented, alongside practical applications that demonstrate the effectiveness of computer simulations in statistical and mathematical modelling. Emphasis is placed on interpreting simulation results and communicating findings through appropriate data visualisation techniques. Throughout the unit, R and/or Python are used as the primary programming environments.

Unit Learning Outcomes

Curtin University's six Graduate Capabilities indicate to employers that graduates possess discipline knowledge and valuable skills. Each course unit addresses these capabilities through specific learning outcomes, which outline what students need to know and do to succeed. Assessments are designed to test these outcomes, ensuring that upon completion, students have met all learning objectives.

Your course has been designed so that on graduating you will have achieved all of Curtin's Graduate Capabilities through the assurance of the learning process in each unit.

	On successful completion of this unit student can:	Graduate Capabilities addressed
1	Explain the scope and limitation of simulation	 
2	Develop simulation procedures in an appropriate computing environment	
3	Analyse and interpret simulation result	
4	Apply simulation techniques in a discipline specific context	

Curtin's Graduate Capabilities

	GC1: Apply knowledge, skills and capabilities		GC2: Innovative, creative and / or entrepreneurial		GC3: Effective communicators with digital competence
	GC4: Globally engaged and responsive		GC5: Culturally competent to engage respectfully with local First Peoples and other diverse cultures		GC6: Industry-connected and career-capable

Find out more about [Curtin's Graduate Capabilities](#).

Learning Activities

All students enrolled in this unit are expected to attend all scheduled weekly learning activities. For online content, students should access live sessions or recorded materials via Blackboard, as appropriate. All students will have access to the following learning activities:

- Lectures (face-to-face, live online, and recorded via iLecture)
 - One two-hour lecture per week.
 - All lectures are recorded and available through the **iLecture** tab on Blackboard.
 - Students are expected to review the weekly materials prior to attending each lecture.
 - If a lecture is missed, students should view the recorded iLecture before attending or attempting the corresponding workshop.
- Workshop (face-to-face and live online via MS Team)
 - One two-hour workshop per week focusing on simulations in **R/RStudio** and problem-solving activities that illustrate concepts introduced in the lectures.
 - Students must bring a personal laptop with **R/RStudio** (and Python/Jupyter Notebook) installed.
 - Workshop exercises are designed to develop a deeper understanding of key concepts, methods, simulation techniques, programming skills, and interpretation relevant to the unit topics.

Learning Resources

Library Reading List

The Reading List for this unit can be accessed through Blackboard.

The text(s) for this unit are:



Carlton, M. A., & Devore, J. L. (2017). Probability with Applications in Engineering, Science, and Technology (J. L. Devore, Ed.; Second edition.). Springer Nature.
<https://doi.org/10.1007/978-3-319-52401-6>

Electronic:No **Essential:**No **Resource Type:** Book **ISBN:** 9783319524016
Url: https://curtin.alma.exlibrisgroup.com/leganto/public/61CUR_INST/citation/19800994200001951?auth=LOCAL



Jones, O. (2014). Introduction to scientific programming and simulation using R (R. Maillardet & A. Robinson, Eds; 2nd ed.). Chapman and Hall/CRC an imprint of Taylor and Francis. <https://doi.org/10.1201/b17079>

Electronic:No **Essential:**No **Resource Type:** Book **ISBN:** 9781040079355;
1040079350; 9780429196195;
04291961
Url: https://curtin.alma.exlibrisgroup.com/leganto/public/61CUR_INST/citation/19800994740001951?auth=LOCAL



Templ, M. (2016). Simulation for data science with R : harness actionable insights from you data with computational statistics and simulations using R (1st edition). Packt Publishing.

Electronic:No **Essential:**No **Resource Type:** Book **ISBN:** 9781785885877;
9781785881169
Url: https://curtin.alma.exlibrisgroup.com/leganto/public/61CUR_INST/citation/19800995320001951?auth=LOCAL



Albert, J. (2007). Bayesian Computation with R [electronic resource] (SpringerLink (Online service), Ed.). Springer New York. <https://doi.org/10.1007/978-0-387-71385-4>

Electronic:No **Essential:**No **Resource Type:** Book **ISBN:** 9780387713854
Url: https://curtin.alma.exlibrisgroup.com/leganto/public/61CUR_INST/citation/19800996440001951?auth=LOCAL

Essential Software



- R and RStudio (required; assessed)
- Python and Jupyter Notebook (optional; not assessed)

Assessment

Assessment policy exemptions

There are no exemptions to the assessment policy

Assessment Schedule

	Task	Value %	Date Due	Unit Learning Outcome(s) Assessed	Late Assessments Accepted?	Assessment Extensions Considered? *
1	Computer-based test	20 %	Week: 5,11 Day: 30 Apr & 19 Mar 2026 Time: 10:00AM AWST	1,2,3	No	Yes
2	Project	30 %	Week: 6,14 Day: 27 Mar & 22 May 2026 Time: 11:59PM AWST	2,3,4	Yes	Yes
3	Final Exam	50 %	Week: 16-17 Day: TBA Time: TBA	1,2,3,4	No	Yes

*Please refer to the Late Assessment and the Assessment Extension sections below for specific details and conditions.

Detailed Information on assessment tasks

Computer-based test

Two **computer-based in-class tests** will be conducted as follows:

- **In-class Test 1** (10%) will be held in **Week 5** and will assess material covered in **Weeks 1-4**.
- **In-class Test 2** (10%) is scheduled in **Week 11**, assessing the materials covered in **Weeks 5-7 and 10**.
- Further details, including format and requirements, will be provided on Blackboard.

Project

The **Individual simulation project** consists of two components:

- **Project Proposal** due at the end of **Week 6**.
- **Final Report** (30%) due at the end of **Week 14**.
- Further details, including submission format, templates, and assessment rubrics, will be provided on Blackboard.

Final Exam

The **final examination** (50%) assesses material covered in **Week 1-14**. Further details will be provided on Blackboard.

Pass requirements

Students must achieve a Final Mark of 50 or greater to pass this unit.

Assessment Moderation

Fair assessment through moderation

Moderation describes a quality assurance process to ensure that assessments are appropriate to the learning outcomes, and that student work is evaluated consistently by assessors. Minimum standards for the moderation of assessments are described in the [Assessment and Student Progression Manual](#).

Pre-marking moderation

- Peer review of the assessment design and assessment task details especially for the Computer based test and the Final Examination;
- Peer review of marking guide for the Project.

Intra-marking / Post-marking moderation

- Moderation of a sample of student work near grade boundaries;
- Moderation of all borderline results to conform standard.

Late Assessment

Where the submission of a late assessment is permitted, late penalties will be consistently applied in this unit.

Where a late assessment **is** permitted for an assessment item or the entirety of the unit (refer to the Assessment Schedule table in this Unit Outline) and the student does not have an approved assessment extension:

1. For assessment items submitted within the first 24 hours after the due date/time, students will be penalised by a deduction of 5% of the total marks allocated for the assessment task;
2. For each additional 24 hour period commenced an additional penalty of 10% of the total marks allocated for the assessment item will be deducted; and
3. Assessment items submitted more than 168 hours late (7 calendar days) will receive a mark of zero.

Where late assessment **is NOT** permitted for an assessment item or the entirety of the unit (refer to the Assessment Schedule table in this Unit Outline) and the student does not have an approved assessment extension:

1. All assessment items submitted after the due date/time will receive a mark of zero.

Assessment Extension

Where an application for an assessment extension **is** permitted for an assessment item(s) within this unit (refer to the Assessment Schedule table in this Unit Outline):

1. A student who is unable to complete an assessment item by/on the due date/time as a result of exceptional circumstances beyond the student's control, may apply for an assessment extension on the [Assessment Extension Form](#) and within the student OASIS (My Studies tab – Study Essentials) account.
2. Submit the application for an Assessment Extension with supporting documentation [via the online form](#).
3. An application may be accepted up to five working days after the due date/time of the assessment item where the student is able to provide a verifiable explanation as to why they were not able to submit the application prior to the assessment due date/time.

Where an application for an assessment extension **is NOT** permitted for an assessment item(s) within this unit (refer to the Assessment Schedule table in this Unit Outline):

1. All assessment items submitted after the due date/time will be subject to late penalties or receive a mark of zero depending on the unit permitting late assessment submissions.

Deferred Assessments

If your results show that you have been granted a deferred assessment you should immediately check OASIS for details.

Deferred examinations/tests will be held from 13/07/2026 to 18/07/2026. Notification to students will be made after the Board of Examiners' meeting via the Official Communications Channel (OCC) in OASIS.

Further Assessments

Further assessments, if granted by the Board of Examiners, will be held between 13/07/2026 to 18/07/2026. Notification to eligible students granted a further assessment will be made after the Board of Examiners meeting via the Official Communications Channel in OASIS.

It is the responsibility of the student to be available to complete the requirements of a further assessment. If your results show that you have been granted a further assessment you should immediately check OASIS for details.

Reasonable adjustments for students with disabilities/health circumstances likely to impact on studies

A [Curtin Access Plan](#) (CAP) is a document that outlines the type and level of support required by a student with a disability or health condition to have equitable access to their studies at Curtin. Carers for people with disability may also be eligible for support. This support can include alternative exam or test arrangements, study materials in accessible formats, access to Curtin's facilities and services or other support as discussed with an advisor from [AccessAbility Services](#).

Documentation is required from your treating Health Professional to confirm your health circumstances or carer responsibilities.

If you think you may be eligible for a CAP, please contact [AccessAbility Services](#). If you already have a CAP, please provide it to the Unit Coordinator in week 1 of each study period.

Referencing style

The referencing style of this unit is APA 7th Ed.

More information can be found on this style from the library web site <https://uniskills.library.curtin.edu.au/referencing/apa7/introduction/>

Privacy

[Curtin's privacy statement](#) describes how personal information is handled. Curtin may record or transmit your image or voice during learning activities or class participation, both on campus and internationally. Students may also record for study purposes but must not share these recordings publicly and must seek permission from those recorded. Recordings cannot be used for commercial purposes or shared beyond personal study. Breaching the [privacy policy or procedures](#) may lead to disciplinary action under [Statute No 10](#). For privacy concerns, please contact your Unit Coordinator.

Copyright

The course material for this unit is provided solely for your personal research and study. It is protected by [copyright](#) and sharing it on third-party websites without Curtin University's written consent is a copyright infringement.

Academic Integrity

Curtin's [Student Charter](#), [Academic Integrity Program \(AIP\)](#), and core [Values](#) guide expectations regarding student behaviour and responsibilities. Information on these topics can be found on the [Academic Integrity Website](#).

Appropriate Use of Generative Artificial Intelligence (Gen-AI) technologies

Curtin supports the philosophy of teaching students to appropriately use Gen-AI technologies in an ethical and responsible way. Gen-AI technology is rapidly evolving and being incorporated into software programs, so it is important to understand how it can and cannot be used within your studies.

Check your assessment instructions carefully before using any Gen-AI software (e.g. Chat GPT, Midjourney, GitHub Copilot, etc.). You are not permitted to use Gen-AI software in any assessment task unless written permission is explicitly granted by the Unit Coordinator (e.g. within Blackboard or the assignment specifications). Where use of Gen-AI is approved, you must use it in accordance with those instructions. Unapproved, inappropriate, or undisclosed use may be dishonest or unfair behaviour, and thus considered misconduct.

Visit the [appropriate use of Gen-AI technologies website](#) and [library website on Gen-AI](#) for more information.

Academic Integrity Warnings

An [Academic Integrity Warning](#) may be issued to a student in limited circumstances and only where misconduct is not involved.

Academic Misconduct

Staff members are required to report suspected misconduct. [Academic misconduct](#) means conduct by a student that is dishonest or unfair in connection with any academic work. This includes all types of plagiarism, cheating, collusion, falsification or fabrication of content, and behaviours like falsifying medical certificates for extension. [Contract cheating](#), the use of file sharing, translation services/apps, paraphrasing tools (text-spinners), article generators, unapproved and inappropriate use of Gen-AI tools, and assignment help websites also may be considered academic misconduct. The longer term personal, social, and financial consequences of misconduct can be severe, so please ask your tutors or unit coordinator if you need clarification or are unsure what to do.

Information and Communications Technology (ICT) Expectations

Curtin students need a reliable internet connection to access OASIS email, the Learning Management System, and [Library Services](#). A computer or mobile device is also required for preparing and submitting assessments.

You may be required to use [Respondus Monitor with Lockdown Browser](#) to [verify your identity](#) and monitor your behaviour during online tests and exams. This software requires a [computer](#) with a webcam, a microphone, and a stable internet connection. It is not compatible with tablets or iPads at Curtin.

For general ICT assistance, visit the [system help and guides website](#). For study resources and assistance, check out the [UniSkills website](#).

Additional information

Enrolment

It is your responsibility to ensure that your enrolment is correct - you can check your enrolment through the eStudent option on OASIS, where you can also print an Enrolment Advice.

Student Rights and Responsibilities

Students must be aware of all relevant legislation, policies, and procedures concerning their rights and responsibilities. This information is available on the [student rights and responsibilities](#) website.

Student Equity

Several factors might hinder students from performing their best in studies or assessments, such as disabilities, medical conditions, significant caring responsibilities, pregnancy, religious practices, remote living, or other reasons. If you believe you are unfairly disadvantaged, contact the appropriate service. University staff can only assist if they are aware of your circumstances, so please reach out for help.

To discuss your needs in relation to:

1. Disability or medical conditions, contact [AccessAbility Services](#)
2. Elite athletes, contact [Elite Athlete Coordinator](#)
3. All other grounds, contact the [Student Wellbeing Advisory Service](#)

Recent Unit Changes & Response to Student Feedback

Students are encouraged to provide feedback through student surveys (such as [Insight](#) and the annual [Student Experience Survey](#)) and interactions with teaching staff.

Listed below are some recent changes to the unit as a result of student feedback.

- An online option is available for workshop activities.
- Python materials are provided in parallel with R, offering students greater flexibility

Program Calendar

Week	Begin Date	Lecture	Pre-reading*	Workshop	Assessment
0	9 Feb	Orientation Week			
1	16 Feb	Introduction	JMR: Ch 14-15 CD: Ch 1	Revision of elementary probability	
2	23 Feb	Statistical Distributions: Univariate & Joint	JMR: Ch 16-17 CD: Ch 4	Computation using Built-in Functions in R/Python	
3	2 Mar	Statistical Inference: frequentist & Bayesian approach	CD: Ch 5.1-5.2, 5.6 JMR: Ch 18	Computation using Built-in Functions in R/Python	
4	9 Mar	Simulation of discrete random variables	JMR: Ch 20.1-.2 CD: Ch 1.6, 2.8	Simulation & Computation using R/Python	
5	16 Mar	Simulation of continuous random variables	JMR: Ch 20.3-20.4 CD: Ch 3.8	Simulation & Computation using R/Python	Test 1 (10%)
6	23 Mar	Simulation of multivariate distribution (Part 1)	CD: Ch 4.10	Simulation & Computation using R/Python	Project Proposal (5%)
7	30 Mar	Simulation of multivariate distribution (Part 2)	CD: Ch 4.10	Simulation & Computation using R/Python	
8	6 Apr	Tuition Free Week			
9	13 Apr				
10	20 Apr	Data Visualisation	Material on BB	Data Visualisation with R/Python	
11	27 Apr	Monte Carlo Integration, simulation and Gibbs Sampling	Material on BB	Simulation & Computation using R/Python	Test 2 (10%)

12	4 May	Simulation in Practice - case studies (Part 1)	
13	11 May	Simulation in Practice - case studies (Part 2)	
14	18 May	Revision and project consultations	Final Report (25%)
15	25 May	Study Week	
16	1 Jun	Centrally Scheduled Examinations	
17	8 Jun		

***JMR:** Jones, O., Maillardet, R., & Robinson, A. (2009). *Introduction to scientific programming and simulation using R*. Chapman and Hall/CRC.

***CD:** Carlton, M. A., & Devore, J. L. (2017). *Probability with applications in engineering, science, and technology*. Berlin: Springer.